

LIFEDESK

Your Personal Problem-Solving Agent

“The more it understands your life, the less you have to explain.”

**ROUND 1
PRODUCT
THESIS**

LifeDesk turns scattered personal context—devices, documents, tasks, expenses, maintenance issues, preferences, and previous solutions—into persistent personal memory. When a familiar problem appears, the agent recalls what happened before and uses that history to give increasingly personalized help.

Hackathon fit	What LifeDesk demonstrates
Innovation (30%)	A personal problem-solving workflow built around compounding memory, not another generic chatbot.
Persistent memory (25%)	Memory is the product mechanism: retain, recall, correct, and learn over time.
Technical (20%)	Agent orchestration, structured entities, hybrid retrieval, evidence, and traceable actions.
UX (15%)	Fast “What happened last time?” flow with timelines, documents, and action cards.
Impact (10%)	Saves time and money by preventing repeated troubleshooting, missed renewals, and duplicate spending.

Prepared as a practical build blueprint. Product concept basis: the supplied LifeDesk recommendation, adapted into the same blueprint structure as the provided ECHOLESS project document.

1. Executive Summary

The complete product idea, in one page

What LifeDesk does

LifeDesk is a single AI agent that remembers a person's recurring problems, preferences, tasks, devices, expenses, maintenance issues, documents, and previous solutions. It uses that history to answer questions, retrieve evidence, suggest next steps, and become more useful with repeated interactions.

The problem

Personal information is fragmented across chats, email, receipts, manuals, notes, calendars, banking records, and memory. When the same issue returns, people often start from zero: they search again, repeat troubleshooting, forget warranty details, or pay for something twice.

The product insight

CORE INSIGHT

Personalization compounds. The valuable memory is not just “what the user said.” It is the connection between an entity, an event, a decision, an outcome, and the evidence that supports it.

Primary user

- Students and professionals managing many devices, documents, subscriptions, and recurring tasks
- Households tracking maintenance, warranties, expenses, and service providers
- Anyone who repeatedly solves the same practical problems

North-star question

“Have I dealt with this before—and what actually worked?” LifeDesk answers with personalized context and evidence rather than generic advice.

2. Problem, Opportunity and Differentiation

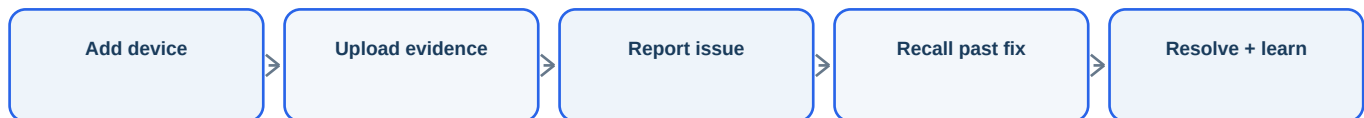
What is already common	What LifeDesk changes
Chatbots with short-lived conversation context	A persistent personal memory layer across interactions and domains.
Cloud folders and document search	Documents connected to devices, expenses, tasks, dates, and past actions.
Reminder apps	Reminders informed by history, warranties, service intervals, and actual usage.
Generic troubleshooting guides	Context-aware guidance based on previous fixes and the user's specific setup.
Expense dashboards	Expense awareness linked to subscriptions, purchases, renewals, and duplicate payments.

3. End-to-End User Journey

Step	User action	Agent behavior	Visible result
1	Add a device	Extracts model, purchase date, warranty, manual and receipt details.	Structured device card
2	Report a problem	Links the issue to the device and recalls similar historical events.	Relevant past fixes
3	Inspect history	Connects issue → diagnosis → action → outcome → evidence.	Problem timeline
4	Ask “what changed?”	Compares the current symptoms/setup with previous cases.	Changed vs. unchanged factors
5	Review recommendation	Generates a concise evidence-backed next step.	Action card + sources
6	Confirm / correct	User accepts, rejects, or edits the memory.	Corrected memory
7	After resolution	Outcome is stored for future use.	Memory becomes richer

Recommended MVP scenario

Start with **devices + maintenance issues + documents**. A user adds a washing machine, uploads a receipt/manual, and later reports a recurring noise. LifeDesk recalls that the same symptom previously came from the drain filter, surfaces the relevant manual section and past service invoice, and checks the warranty status.



DEMO RESULT

“This washing machine had the same noise 8 months ago. The previous service note says the drain filter was cleaned. Your warranty is still active, and the receipt is attached. Here are the steps and the service contact.”

Personalization loop

Each useful interaction creates a new link between **entity** → **event** → **action** → **outcome**. Future answers can therefore become more specific without requiring the user to repeat the full story.

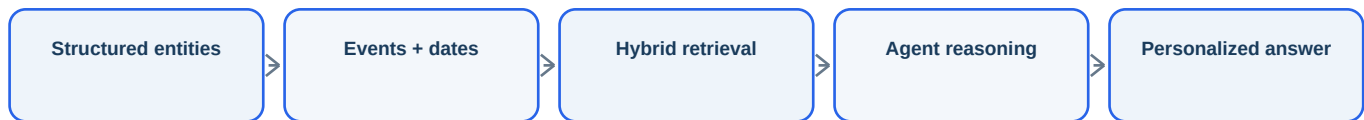
4. How Persistent Memory Becomes the Core of LifeDesk

LifeDesk should treat memory as structured experience, not just a chat transcript.

Memory operation	LifeDesk implementation
Retain	Store structured experiences from devices, documents, purchases, problems, fixes, tasks, expenses, preferences, and outcomes.
Recall	Retrieve relevant memories using semantic search plus explicit filters such as device, date, amount, provider, document type, or status.
Reflect	Reason across retrieved experiences to identify recurring issues, useful preferences, repeated spending, upcoming deadlines, and stale information.
Correct	Let the user edit or invalidate a memory when a device is replaced, a subscription is cancelled, or a preference changes.

Why structured memory beats a plain vector store

Questions such as “What did I spend on the car this year?” require real dates, amounts, and entity relationships. Questions such as “How did I fix this noise last time?” benefit from semantic retrieval. LifeDesk therefore combines structured queries with semantic search instead of treating every memory as an unstructured paragraph.



Memory is not a decoration

Without persistent memory, LifeDesk becomes a generic assistant. With memory, the same question can produce a response grounded in the user's own history, documents, spending, devices, and previous outcomes.

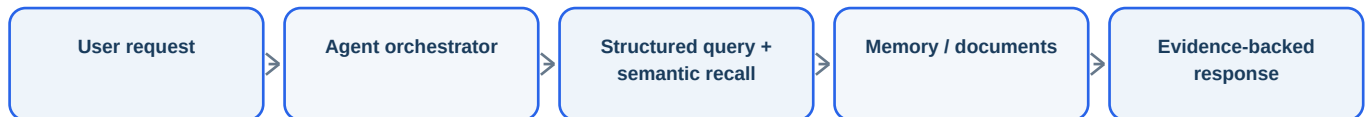
TRUST PRINCIPLE

A remembered fact should carry its source, timestamp, and confidence where possible. If evidence conflicts or is stale, LifeDesk should ask or surface the conflict instead of silently guessing.

5. Recommended Technical Architecture

Layer	Recommended technology	Purpose
Frontend	React / Next.js + Tailwind	Personal dashboard, memory timeline, document cards, issue history and action UI.
Agent API	Python + FastAPI	Request routing, tool calls, validation, structured outputs and business logic.
Memory layer	Hindsight Cloud or self-hosted Hindsight	Persistent retain / recall / reflect workflow for LifeDesk memories.
Model layer	Microsoft Foundry model endpoint / Azure-hosted model	Extraction, reasoning, summarization and response generation.
Structured data	PostgreSQL / Cosmos DB / JSON for MVP	Entities, dates, amounts, status, relationships and metadata.
Documents	Azure Blob Storage or local object store	Receipts, manuals, invoices, warranties and other evidence.
Runtime	Azure Container Apps or App Service	Simple deployment path for a hackathon demo.
Observability	Application Insights	Latency, failures, traces and agent behavior.
Identity	Microsoft Entra ID (optional)	Authentication and scoped access for a fuller version.

Architecture flow



Core request flow

POST /analyze → classify request → identify entities → retrieve structured facts + semantic memories → inspect evidence → reason → return structured JSON for UI

POST /learn → capture correction/outcome → update entity state → retain new experience → improve future recall context

6. Core Product Modules

Module	What it does
A. Personal Intake	Accept a question, device, receipt, document, task, expense or issue.
B. Entity Memory	Maintain first-class records for devices, documents, tasks, expenses, providers, people and subscriptions.
C. Memory Matcher	Retrieve related historical events using relevance, recency, entity links and outcome similarity.
D. Problem Replay	Reconstruct what happened previously: symptom → diagnosis → action → outcome.
E. What-Changed Engine	Compare current conditions with previous cases across model, timing, symptoms, settings and context.
F. Document Retriever	Find the exact receipt, manual, warranty or invoice supporting the answer.
G. Personalization Engine	Apply stored preferences such as budget limits, preferred brands and advice style.
H. Learning Capture	Store confirmed outcomes and corrections so future answers improve.

7. Memory and Data Model

Example memory record

Field	Example
event_type	maintenance_issue
entity	Washing Machine — LG 8kg
timestamp	2026-04-18
issue	Loud grinding noise during spin cycle
previous_fix	Cleaned drain filter and removed trapped object
outcome	Noise stopped after service
evidence	service-invoice-2026-0418.pdf
warranty_status	Active until 2027-02-11
preferences	Prefer low-cost fixes before service calls

What to capture

Entity identity • dates • amounts • source documents • actions • outcomes • status • relationships • user corrections • confidence / freshness.

8. UI / UX Plan — Make It Look Like a Real Product

Screen	Design
Screen 1: LifeDesk Home	Top status: Memory Health, saved items, upcoming deadlines. Primary CTA: ASK LIFEDESK. Cards for active issues, expiring warranties and recent memories.
Screen 2: Problem Analysis	Left: current issue. Center: connected historical timeline. Right: evidence, previous fixes, changed/unchanged factors.
Screen 3: Memory Replay	Timeline: problem → diagnosis → action → outcome → evidence. Tap a node to open the underlying document or record.
Screen 4: Personal Knowledge	Searchable view of devices, documents, expenses, tasks and preferences, with edit/correct controls.
Screen 5: Learning Curve	Interaction 1: generic help. Interaction 5: relevant history. Interaction 15: recurring pattern. Interaction 20: proactive personalized help.

Primary visual hierarchy



VISUAL RULE

Avoid a generic chat-only layout. Chat can be a secondary interaction. The primary visual should be a personal memory timeline, entity cards, evidence panels, and clear action buttons.

Example action card

LIFEDESK RECALL	WHY IT MATTERS	ACTION
Same washing-machine noise found in 1 previous service record.	The earlier fix was successful and the warranty is still active.	Check drain filter → open manual → contact service if unresolved.

9. The 60-Second Round 1 Demo

Time	What you show	One-line narration
0–8s	Recurring washing-machine problem	"This happened before. Let's see what LifeDesk remembers."
8–18s	Click ASK LIFEDESK / issue check	"LifeDesk searches my personal memory, not just the open web."
18–30s	Past service memory appears	"It found the same device, a previous issue, and the fix that worked."
30–42s	Open Memory Replay	"Here is what happened, when it happened, and the evidence."
42–52s	Open What Changed	"The symptoms match, but the warranty status and device context are different."
52–60s	Action + evidence card	"LifeDesk turns my history into a personalized next step and remembers the outcome."

The demo story



DEMO PAYOFF

The audience should immediately understand the difference between a generic assistant and a personal agent: the generic assistant explains what might work; LifeDesk can show what worked for this person, on this device, with this evidence.

Second demo option

Expense awareness: upload a subscription receipt, ask "Have I paid for this twice?", retrieve matching transactions, show the dates and amounts, and create a renewal reminder. This demonstrates that the same memory architecture can grow beyond maintenance.

10. 8-Hour Hackathon Build Plan

Phase	Target output	Priority
Hour 0–1	Project skeleton, Hindsight connection, environment variables	Must have
Hour 1–2	Seed synthetic LifeDesk corpus: 10–20 devices, issues, receipts and documents	Must have
Hour 2–3	Implement retain + recall + structured agent response	Must have
Hour 3–4	Implement What-Changed comparison and evidence linking	Must have
Hour 4–5	Build polished dashboard, entity cards and timeline	Must have
Hour 5–6	Add correction controls + learning capture	High value
Hour 6–7	Demo script, edge cases, empty states, seed refresh	High value
Hour 7–8	Deploy, rehearse, record screenshots/video, README	Must have

Minimum viable product

MVP element	Scope
One persona	Student / professional managing personal devices and documents
One workflow	Recurring problem → recall previous solution → evidence-backed action
One memory type	Device + maintenance experience
One killer action	Replay previous fix + compare current situation
One polished dashboard	Home + issue analysis + memory timeline

Do NOT build in the first version

• Full bank integration • Complex household sharing • Full enterprise RBAC • Large-scale email ingestion • Mobile app • Multi-agent orchestration • Automatic actions that change external systems without confirmation

COLD-START STRATEGY

Make ingestion effortless: photograph receipts, upload a manual, paste a service message, or add a device in under a minute. The first successful recall should happen quickly so the user sees why memory matters.

11. Trust, Safety and Edge Cases

Risk	Mitigation
Wrong memory	Show the source and date; allow the user to correct or invalidate the memory.
Stale information	Attach freshness dates and surface when a warranty, address, device or subscription may have changed.
Contradictory records	Show both records with timestamps and ask the user to confirm the current state.
False similarity	Explain why a past issue was matched; never claim two cases are identical from weak evidence.
Missing evidence	Say what is unknown and ask for the missing receipt, document or detail.
Sensitive documents	Use scoped access, encryption, retention controls and deletion controls in real deployment.
Overconfident advice	Frame recommendations as evidence-based assistance; require confirmation before consequential actions.
Cold start	Provide guided setup and easy ingestion so the memory base grows quickly.

IMPORTANT PRODUCT PRINCIPLE

LifeDesk should never pretend to remember something it does not have evidence for. A trustworthy answer can be: "I don't have enough reliable history yet. Here is what I do know."

12. How LifeDesk Maps to the Hackathon Rubric

Judging area	How the project answers it	Demo proof
Innovation	Personal problem-solving agent with compounding memory	Recurring issue becomes easier the second time
Memory	Persistent structured + semantic personal context	Memory timeline + learning curve
Technical	Agent + structured entities + hybrid retrieval + evidence	Architecture + traces
UX	One clear question-to-action workflow	60-second problem replay
Real-world impact	Saves time, avoids repeated troubleshooting and missed renewals	Device, document and expense examples

13. Product Roadmap After the Hackathon

Stage	Expansion
V1 — Hackathon	Synthetic data, devices + maintenance, document recall, problem replay, persistent memory.
V2 — Personal pilot	Email/receipt ingestion, calendar tasks, subscriptions, richer preferences, user corrections.
V3 — Connected life	Optional bank data, smart-home/device integrations, service-provider records, household sharing.
V4 — Personal learning system	Recurring-issue watchlists, proactive reminders, spending anomalies, cross-domain pattern detection and outcome feedback.

Potential usages

• Device troubleshooting and maintenance • Warranty tracking • Document retrieval • Subscription renewals • Expense awareness • Recurring household tasks • Vehicle maintenance • Travel/document preparation • Personal project support

14. Final Pitch, Naming and Submission Language

ONE-LINE PITCH

LifeDesk is a personal AI problem-solving agent that remembers your devices, documents, tasks, expenses, preferences and previous solutions—so the next time a problem appears, it already knows your context.

30-second explanation

Most assistants help with the problem in front of you. LifeDesk remembers the problems you have already solved. It connects devices, documents, expenses, tasks and past outcomes, then uses that history to give increasingly personalized help. The goal is simple: stop starting from zero.

Project positioning

Category	Personal AI Agent / Personal Memory
Primary technology	Persistent structured memory + semantic retrieval + agent orchestration
Core differentiator	Compounding personalization across recurring real-life problems
Primary demo	A recurring problem is solved faster because LifeDesk remembers the previous solution
Tagline	"Your life, remembered. Your problems, solved faster."

15. Final Build Principle

DEFINITION OF DONE

A polished agent that does one workflow extremely well is more convincing than a huge platform with unfinished features. Build the memory loop first: **capture** → **recall** → **compare** → **act** → **confirm** → **learn**.

Suggested repository structure

lifedesk/ • apps/web/ • services/agent/ • memory/ • data/seed/ • prompts/ • docs/ • .env.example • README.md

Core request flow

POST /analyze → identify entity → recall memories → retrieve evidence → compare context → generate structured response → display action + sources
 POST /learn → store outcome/correction → update entity → retain experience → improve future recall

LifeDesk turns personal history into practical help. The memory is not the feature around the product—the memory is what makes the product increasingly useful.